



PREMIER UNIVERSITY CHATTOGRAM

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

LAB REPORT

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REPORT NAME	YOUTUBE TRENDING VIDEO DATA ANALYSIS USING HADOOP MAPREDUCE AND APACHE PIG.	
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Problem Statement

Analyze a YouTube trending video dataset using Hadoop MapReduce and Apache Pig based on different data fields to produce meaningful insights. The project implements three MapReduce analyses and five Pig analyses on a cleaned YouTube trending dataset stored in HDFS.

The analysis focuses on video categories, video views, video likes, and channel-level video counts. The project also demonstrates preprocessing, HDFS storage, MapReduce execution on YARN, and Pig execution in MapReduce mode.

```
C:\>cd hadoop
C:\hadoop>start-yarn.cmd
starting yarn daemons
C:\hadoop>start-dfs.cmd
C:\hadoop>jps
15760 NodeManager
15252 Jps
17652 ResourceManager
1932 NameNode
8268 DataNode
C:\hadoop>
```

Dataset Description

Dataset Source:

<https://www.kaggle.com/datasets/bsthere/youtube-trending-videos-stats-2026>

Dataset used for the current analysis: US_Trending.csv

Local dataset folder: C:\Final

The downloaded dataset collection contains country-specific trending video CSV files. For this project, US_Trending.csv was selected as the analysis input. The preprocessing step produced US_Trending_Clean.tsv, which is used as the common HDFS input for all MapReduce and Pig operations.

Column	Description
video_id	Identifier of a YouTube video.
trending_date	Date associated with the trending record.
title	Title of the video.
channel_title	Displayed name of the YouTube channel.
views	Number of views recorded for the video.
likes	Number of likes recorded for the video.
dislikes	Dislikes field available in the downloaded dataset.
publish_time	Video publication timestamp.
category_id	Numeric YouTube category identifier.
tags	Tags associated with the video.
comments	Number of comments recorded for the video.
channel_id	Identifier of the YouTube channel.
description	Video description text.

Project Environment

Component	Version / Value
Operating Environment	Windows
Java	1.8.0_411
Hadoop	3.2.4
Apache Pig	0.18.0
HDFS Input	/user/DELL/youtube_project/clean/US_Trending_Clean.tsv
Pig Execution Mode	MapReduce mode (-x mapreduce)

Data Preprocessing Summary

The raw CSV contains text fields such as title, tags, and description. A Java preprocessing program using Apache Commons CSV was used to parse the CSV safely, normalize tab/newline characters, retain the fields required by the analyses, check repeated video IDs, and write a tab-separated clean file for Hadoop and Pig.

Item	Value
Raw input file	C:\Final\US_Trending.csv
Clean output file	C:\Final\US_Trending_Clean.tsv
Total raw rows	16,400
Skipped rows	0
Unique videos kept	16,400
Clean file size	2,129,302 bytes

```

=====
Preprocessing completed
=====
Total raw rows      : 16400
Skipped rows       : 0
Unique videos kept  : 16400
Output file        : C:\Final\US_Trending_Clean.tsv
=====

```

```

C:\FinalProject\Preprocessing>dir C:\Final\US_Trending_Clean.tsv
Volume in drive C is SSD
Volume Serial Number is 16B2-D78D

Directory of C:\Final

09/10/2026  04:17 PM                2,129,302 US_Trending_Clean.tsv
               1 File(s)                2,129,302 bytes
               0 Dir(s) 122,709,610,496 bytes free

```

Summary of Data Analysis

Following MapReduce analyses are implemented:

1. Top 10 Categories by Video Count
2. Top 10 Most Viewed YouTube Videos
3. Top 10 Channels by Number of Videos

Following Pig analyses are implemented:

1. Calculate Top 5 Categories of YouTube Videos
2. Calculate Top 10 Liked YouTube Videos
3. Calculate Top 10 Liked Videos by Categories
4. Calculate Top 10 Viewed YouTube Videos
5. Calculate Top 10 Viewed Videos by Categories

The selected dataset does not contain a rating field; therefore, likes are used as the engagement metric for the corresponding Pig analyses.

The complete project source code, Pig scripts, processed dataset, and result files are maintained in a public GitHub repository. Repository details are provided near the end of this report.

MapReduce Analysis

The MapReduce portion contains three analyses. The cleaned TSV file is stored in HDFS and used as the common input for all three MapReduce programs.

Data Preprocessing and HDFS Loading

Purpose: Convert the raw CSV into a predictable tab-separated input format that can be parsed reliably by both Java MapReduce and Pig scripts.

Code

```
// Core preprocessing logic
CSVParser parser = CSVFormat.DEFAULT.builder()
    .setHeader()
    .setSkipHeaderRecord(true)
    .setIgnoreEmptyLines(true)
    .build()
    .parse(reader);

for (CSVRecord record : parser) {
    VideoRecord current = new VideoRecord(record);
    if (current.videoId.isEmpty()) {
        skippedRows++;
        continue;
    }
    VideoRecord existing = latestVideos.get(current.videoId);
    if (existing == null ||
        parseTrendingDate(current.trendingDate)
            .isAfter(parseTrendingDate(existing.trendingDate))) {
        latestVideos.put(current.videoId, current);
    }
}
```

Execution

```
mvn clean compile
mvn exec:java -Dexec.args="C:\Final\US_Trending.csv C:\Final\US_Trending_Clean.tsv"
hdfs dfs -mkdir -p /user/DELL/youtube_project/clean
hdfs dfs -put C:\Final\US_Trending_Clean.tsv /user/DELL/youtube_project/clean/
```

```
Preprocessing completed
=====
Total raw rows      : 16400
Skipped rows       : 0
Unique videos kept  : 16400
Output file        : C:\Final\US_Trending_Clean.tsv
=====
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 3.846 s
[INFO] Finished at: 2026-09-10T16:17:53+06:00
[INFO] -----
```

1) Top 10 Categories by Video Count

Returns the ten category IDs having the highest number of videos in the cleaned dataset. The implementation uses one MapReduce job to count videos by category and a second job to produce the global Top 10 ranking.

This analysis comprises of:

1. Category Mapper: emits category_id and 1 for each valid video record.
2. Category Reducer: sums the occurrences of each category_id.

3. Ranking Mapper: reads category totals produced by the first job.
4. Top 10 Reducer: sorts category totals in descending order and writes the ten highest values.
5. Driver: configures and executes the two jobs sequentially.

Mapper / Reducer Core Logic

```
// Mapper
String[] fields = line.split("\t", -1);
String categoryId = fields[6].trim();
if (!categoryId.isEmpty()) {
    context.write(new Text(categoryId), new IntWritable(1));
}

// Reducer
int sum = 0;
for (IntWritable value : values) {
    sum += value.get();
}
context.write(key, new IntWritable(sum));
```

Execution

```
hadoop jar target\top10-categories-1.0.jar ^
/user/DELL/youtube_project/clean/US_Trending_Clean.tsv ^
/user/DELL/youtube_project/output/category_counts ^
/user/DELL/youtube_project/output/top10_categories
```

```
26-09-10 16:22:28,005 INFO mapreduce.JobSubmitter: number of splits:1
26-09-10 16:22:28,055 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789035126809_0002
26-09-10 16:22:28,055 INFO mapreduce.JobSubmitter: Executing with tokens: []
26-09-10 16:22:28,295 INFO impl.YarnClientImpl: Submitted application application_1789035126809_0002
26-09-10 16:22:28,301 INFO mapreduce.Job: The url to track the job: http://DESKTOP-46P64I0:8088/proxy/application_1789035126809_0002/
26-09-10 16:22:28,301 INFO mapreduce.Job: Running job: job_1789035126809_0002
26-09-10 16:22:39,539 INFO mapreduce.Job: Job job_1789035126809_0002 running in uber mode : false
26-09-10 16:22:39,539 INFO mapreduce.Job:  map 0% reduce 0%
26-09-10 16:22:44,716 INFO mapreduce.Job:  map 100% reduce 0%
26-09-10 16:22:49,783 INFO mapreduce.Job:  map 100% reduce 100%
26-09-10 16:22:49,793 INFO mapreduce.Job: Job job_1789035126809_0002 completed successfully
26-09-10 16:22:49,830 INFO mapreduce.Job: Counters: 54
```

Output

HDFS output: /user/DELL/youtube_project/output/top10_categories/part-r-00000

Local text output: C:\FinalProject\Top10Categories\top10_categories.txt

```
C:\FinalProject\Top10Categories>hdfs dfs -cat /user/DELL/youtube_project/output/top10_categories/part-r-00000
20      4832
10      4561
24      3272
22      1812
1       745
17      613
25      200
28      84
2       73
23      73
```

2) Top 10 Most Viewed YouTube Videos

Returns the ten videos with the highest views values. Each mapper keeps its local Top 10 candidates, and a single reducer compares the mapper candidates and writes the global Top 10 list in descending order of views.

This analysis comprises of:

1. Viewed Mapper: parses video_id, title, channel_title, and views, then keeps the mapper-level Top 10 candidates.
2. Top 10 Reducer: receives candidate videos, sorts them by views in descending order, and writes the global Top 10.
3. Driver: configures one reducer to ensure a single global Top 10 result.

Mapper / Reducer Core Logic

```
// Mapper record extraction
String[] fields = line.split("\t", -1);
String videoId = fields[0].trim();
String title = fields[2].trim();
String channelTitle = fields[3].trim();
long views = Long.parseLong(fields[4].trim());
topVideos.add(new VideoInfo(videoId, title, channelTitle, views));

// Final ranking
allCandidates.sort((a, b) -> {
    int cmp = Long.compare(b.views, a.views);
    return (cmp != 0) ? cmp : a.videoId.compareTo(b.videoId);
});
```

Execution

```
hadoop jar target\top10-viewed-videos-1.0.jar ^
/user/DELL/youtube_project/clean/US_Trending_Clean.tsv ^
/user/DELL/youtube_project/output/top10_viewed
```

```
026-09-10 16:27:40,160 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789035126809_0003
026-09-10 16:27:40,162 INFO mapreduce.JobSubmitter: Executing with tokens: []
026-09-10 16:27:40,399 INFO conf.Configuration: resource-types.xml not found
026-09-10 16:27:40,400 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
026-09-10 16:27:40,490 INFO impl.YarnClientImpl: Submitted application application_1789035126809_0003
026-09-10 16:27:40,542 INFO mapreduce.Job: The url to track the job: http://DESKTOP-46P6410:8088/proxy/application_1789035126809_0003/
026-09-10 16:27:40,543 INFO mapreduce.Job: Running job: job_1789035126809_0003
026-09-10 16:27:49,779 INFO mapreduce.Job: Job job_1789035126809_0003 running in uber mode : false
026-09-10 16:27:49,784 INFO mapreduce.Job: map 0% reduce 0%
026-09-10 16:27:54,879 INFO mapreduce.Job: map 100% reduce 0%
026-09-10 16:28:00,969 INFO mapreduce.Job: map 100% reduce 100%
026-09-10 16:28:01,984 INFO mapreduce.Job: Job job_1789035126809_0003 completed successfully
026-09-10 16:28:02,071 INFO mapreduce.Job: Counters: 54
File System Counters:
```

Output

Output fields: video_id, title, channel_title, views

hdfs dfs -cat /user/DELL/youtube_project/output/top10_viewed/part-r-00000

Local text output: C:\FinalProject\Top10ViewedVideos\top10_viewed_videos.txt

```
C:\FinalProject\Top10ViewedVideos>hdfs dfs -cat /user/DELL/youtube_project/output/top10_viewed/part-r-00000
HyHNuVaZl-k Gorillaz - Feel Good Inc. (Official Video) Gorillaz 1105355210
1V_xRb0x9aw Gorillaz - Clint Eastwood (Official Video) Gorillaz 1014925454
qdsTUFdTEhQ T.I. - No Mediocore ft. Iggy Azalea TIVEVO 420729382
etfIdtm-OC8 T.I. - About The Money ft. Young Thug TIVEVO 387242997
uAOR6ib95kQ Gorillaz - DARE (Official Video) Gorillaz 367587921
koVHN6eO4Xg T.I. - Live Your Life ft. Rihanna [Official Video] T.I. 363500509
CaryjOdYFa0 Brawl Stars: No Time to Explain Brawl Stars 318547842
04mfKJwDSzI Gorillaz - On Melancholy Hill (Official Video) Gorillaz 300678427
nQJACVmankY T.I. - Whatever You Like [Official Video] T.I. 274160249
0cOsLTM519c Brawl Stars Animation: Piper's Sugar & Spice! Brawl Stars 258757017
```

```
C:\FinalProject\Top10ViewedVideos>type C:\FinalProject\Top10ViewedVideos\top10_viewed_videos.txt
video_id  title  channel_title  views
HyHNuVaZJ-k  Gorillaz - Feel Good Inc. (Official Video)  Gorillaz  1105355210
1V_xRb0x9aw  Gorillaz - Clint Eastwood (Official Video)  Gorillaz  1014925454
qdsTUFdTEhQ  T.I. - No Mediocore ft. Iggy Azalea  TIVEVO  420729382
etfIdtm-OC8  T.I. - About The Money ft. Young Thug  TIVEVO  387242997
uAOR6ib95kQ  Gorillaz - DARE (Official Video)  Gorillaz  367587921
koVHN6e04Xg  T.I. - Live Your Life ft. Rihanna [Official Video]  T.I.  363500509
Caryj0dYFa0  Brawl Stars: No Time to Explain Brawl Stars  318547842
04mfKJWDSzI  Gorillaz - On Melancholy Hill (Official Video)  Gorillaz  300678427
nQJACVmankY  T.I. - Whatever You Like [Official Video]  T.I.  274160249
0c0sLTM519c  Brawl Stars Animation: Piper's Sugar & Spice!  Brawl Stars  258757017
```

3) Top 10 Channels by Number of Videos

The combination of channel_id and channel_title is used as the grouping key so that both the channel identifier and readable channel name are retained in the output.

This analysis comprises of:

1. Channel Mapper: emits channel_id + channel_title with a value of 1 for each video.
2. Channel Reducer: counts the total videos associated with each channel key.
3. Ranking Mapper: reads the channel count output.
4. Top 10 Reducer: sorts channel totals in descending order and writes the ten highest values.
5. Driver: executes the counting job and ranking job sequentially.

Mapper / Reducer Core Logic

```
// Mapper
String channelTitle = fields[3].trim();
String channelId = fields[8].trim();
if (!channelId.isEmpty()) {
    context.write(new Text(channelId + "\t" + channelTitle),
        new IntWritable(1));
}

// Reducer
int sum = 0;
for (IntWritable value : values) {
    sum += value.get();
}
context.write(key, new IntWritable(sum));
```

Execution

```
hadoop jar target\top10-channels-1.0.jar ^
/user/DELL/youtube_project/clean/US_Trending_Clean.tsv ^
/user/DELL/youtube_project/output/channel_counts ^
/user/DELL/youtube_project/output/top10_channels
```

```
2026-09-10 16:30:14,950 INFO mapreduce.Job: Running job: job_1789035126809_0005
2026-09-10 16:30:27,157 INFO mapreduce.Job: Job job_1789035126809_0005 running in uber mode : false
2026-09-10 16:30:27,158 INFO mapreduce.Job: map 0% reduce 0%
2026-09-10 16:30:32,240 INFO mapreduce.Job: map 100% reduce 0%
2026-09-10 16:30:37,306 INFO mapreduce.Job: map 100% reduce 100%
2026-09-10 16:30:37,315 INFO mapreduce.Job: Job job_1789035126809_0005 completed successfully
2026-09-10 16:30:37,345 INFO mapreduce.Job: Counters: 54
File System Counters:
```

Output

Output fields: channel_id, channel_title, video_count

hdfs dfs -cat /user/DELL/youtube_project/output/top10_channels/part-r-00000

Local text output: C:\FinalProject\Top10Channels\top10_channels.txt

```
C:\FinalProject\Top10Channels>hdfs dfs -cat /user/DELL/youtube_project/output/top10_channels/part-r-00000
UC63anZxfVGHUEmfBAf5w7pw      CaseOh      500
UC68nKdrLbLL0Vj7iLVkLmmg      Saregama Telugu      500
UC6pGDc4bFGD1_36IKv3FnYg      Crunchyroll      500
UC7_YxT-KID8kRbqZo7MyscQ      Markiplier      500
UCDiFRMQWpcp8_KD4vwIVicw      Emergency Awesome      500
UCGHBUXjDceiIXNdKR0HUZnA      TJR      500
UCLRlryMfL8ffxzrtqv0_k_w      KinoCheck.com      500
UCLkAepWjdyImXSltoFVsYQ      BANGTANTV      500
UCMNEVbszv8ZyvSXoTn3yhpQ      TheBurntPeanut      500
UCMwePVHRpDdfeUcwtDZu2Dw      Monstafluff Music      500
```

```
C:\FinalProject\Top10Channels>type C:\FinalProject\Top10Channels\top10_channels.txt
channel_idchannel_titlevideo_count
UC63anZxfVGHUEmfBAf5w7pw      CaseOh      500
UC68nKdrLbLL0Vj7iLVkLmmg      Saregama Telugu      500
UC6pGDc4bFGD1_36IKv3FnYg      Crunchyroll      500
UC7_YxT-KID8kRbqZo7MyscQ      Markiplier      500
UCDiFRMQWpcp8_KD4vwIVicw      Emergency Awesome      500
UCGHBUXjDceiIXNdKR0HUZnA      TJR      500
UCLRlryMfL8ffxzrtqv0_k_w      KinoCheck.com      500
UCLkAepWjdyImXSltoFVsYQ      BANGTANTV      500
UCMNEVbszv8ZyvSXoTn3yhpQ      TheBurntPeanut      500
UCMwePVHRpDdfeUcwtDZu2Dw      Monstafluff Music      500
```

Pig Analysis

Apache Pig 0.18.0 is executed in MapReduce mode on the same cleaned HDFS input used by the Java MapReduce programs. A JobHistoryServer is started before Pig execution so completed YARN jobs can be queried correctly.

```
set JAVA_HOME=C:\JAVA\jdk-1.8
mapred historyserver
pig -x mapreduce <script-file>
```

1) Top 5 Categories

Returns the Top 5 category IDs based on the number of videos.

Script

```
records = LOAD '/user/DELL/youtube_project/clean/US_Trending_Clean.tsv'
USING PigStorage('\t')
AS (video_id:chararray, trending_date:chararray, title:chararray,
    channel_title:chararray, views:long, likes:long,
    category_id:chararray, comments:long, channel_id:chararray);

data = FILTER records BY video_id != 'video_id';
grouped = GROUP data BY category_id;
counts = FOREACH grouped GENERATE group AS category_id, COUNT(data) AS video_count;
sorted = ORDER counts BY video_count DESC;
top5 = LIMIT sorted 5;
DUMP top5;
STORE top5 INTO '/user/DELL/youtube_project/output/pig/top5_categories'
USING PigStorage('\t');
```

Execution

```
pig -x mapreduce C:\FinalProject\PigAnalysis\scripts\top5Categories.pig
```

```
ecting to job history server
026-09-10 16:48:17,340 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
026-09-10 16:48:17,344 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationSt
ecting to job history server
026-09-10 16:48:17,356 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
026-09-10 16:48:17,359 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationSt
ecting to job history server
026-09-10 16:48:17,377 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Encountered Warnin
'E_CONVERSION_FAILED 3 time(s).
026-09-10 16:48:17,378 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
026-09-10 16:48:17,392 [main] INFO org.apache.pig.Main - Pig script completed in 3 minutes, 33 seconds and 962 milliseconds (213962 ms)
```

Output

hdfs dfs -cat /user/DELL/youtube_project/output/pig/top5_categories/part-r-00000

Local text output: C:\FinalProject\PigAnalysis\outputFiles\top5_categories.txt

```
C:\Windows\System32>hdfs dfs -cat /user/DELL/youtube_project/output/pig/top5_categories/part-r-00000
20      4832
10      4561
24      3272
22      1812
1        745
```

```
C:\Windows\System32>type C:\FinalProject\PigAnalysis\outputFiles\top5_categories.txt
category_id    video_count
20             4832
10             4561
24             3272
22             1812
1               745
```

2) Top 10 Liked YouTube Videos

Returns the ten videos with the highest number of likes in descending order.

Script

```
records = LOAD '/user/DELL/youtube_project/clean/US_Trending_Clean.tsv'
USING PigStorage('\t')
AS (video_id:chararray, trending_date:chararray, title:chararray,
    channel_title:chararray, views:long, likes:long,
    category_id:chararray, comments:long, channel_id:chararray);

data = FILTER records BY video_id != 'video_id';
selected = FOREACH data GENERATE video_id, title, channel_title, likes;
sorted = ORDER selected BY likes DESC;
top10 = LIMIT sorted 10;
DUMP top10;
STORE top10 INTO '/user/DELL/youtube_project/output/pig/top10_liked'
USING PigStorage('\t');
```

Execution

```
pig -x mapreduce C:\FinalProject\PigAnalysis\scripts\top10Liked.pig
```

```

026-09-10 16:58:02,330 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED.
Redirecting to job history server
026-09-10 16:58:02,346 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
026-09-10 16:58:02,349 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED.
Redirecting to job history server
026-09-10 16:58:02,364 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Encountered Warning FIELD_DISCARDED
TYPE_CONVERSION_FAILED 1 time(s).
026-09-10 16:58:02,364 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
026-09-10 16:58:02,379 [main] INFO org.apache.pig.Main - Pig script completed in 3 minutes, 21 seconds and 233 milliseconds (201233 ms)

```

Output

Output fields: video_id, title, channel_title, likes

Local text output: C:\FinalProject\PigAnalysis\outputFiles\top10_liked.txt

`hdfs dfs -cat /user/DELL/youtube_project/output/pig/top10_liked/part-r-00000`

```

C:\Windows\System32>type C:\FinalProject\PigAnalysis\outputFiles\top10_liked.txt
video_id    title      channel_title  likes
HyHnuVaZJ-k Gorillaz - Feel Good Inc. (Official Video) Gorillaz      8968014
1V_xRb0x9aw Gorillaz - Clint Eastwood (Official Video) Gorillaz      8863449
uAOR6ib95kQ Gorillaz - DARE (Official Video) Gorillaz      2605602
04mFKWDSzI Gorillaz - On Melancholy Hill (Official Video) Gorillaz      2506763
qdsTUFDTEHQ T.I. - No Mediocre ft. Iggy Azalea TIVEVO       2246926
ESyFcdPAgV0 Gorillaz - Humility (Official Video) Gorillaz      2164709
koVHN6e04Xg T.I. - Live Your Life ft. Rihanna [Official Video] T.I.         2055870
WXR-bCF5dbM Gorillaz - 19-2000 (Official Video) Gorillaz      2047456
vGBZC9pZHQ Baby Keem, Kendrick Lamar - family ties (Official Video) BabyKeemVEVO 1933399
1SD_L-xic9o Laufey - From The Start (Official Music Video) Laufey      1918312

```

```

C:\Windows\System32>hdfs dfs -cat /user/DELL/youtube_project/output/pig/top10_liked/part-r-00000
HyHnuVaZJ-k Gorillaz - Feel Good Inc. (Official Video) Gorillaz      8968014
1V_xRb0x9aw Gorillaz - Clint Eastwood (Official Video) Gorillaz      8863449
uAOR6ib95kQ Gorillaz - DARE (Official Video) Gorillaz      2605602
04mFKWDSzI Gorillaz - On Melancholy Hill (Official Video) Gorillaz      2506763
qdsTUFDTEHQ T.I. - No Mediocre ft. Iggy Azalea TIVEVO       2246926
ESyFcdPAgV0 Gorillaz - Humility (Official Video) Gorillaz      2164709
koVHN6e04Xg T.I. - Live Your Life ft. Rihanna [Official Video] T.I.         2055870
WXR-bCF5dbM Gorillaz - 19-2000 (Official Video) Gorillaz      2047456
vGBZC9pZHQ Baby Keem, Kendrick Lamar - family ties (Official Video) BabyKeemVEVO 1933399
1SD_L-xic9o Laufey - From The Start (Official Music Video) Laufey      1918312

```

3) Top 10 Liked Videos by Categories

Returns up to ten highest-liked videos for each category. The data is grouped by category_id; records inside each category are sorted by likes in descending order and limited to ten.

Script

```

records = LOAD '/user/DELL/youtube_project/clean/US_Trending_Clean.tsv'
USING PigStorage('\t')
AS (video_id:chararray, trending_date:chararray, title:chararray,
    channel_title:chararray, views:long, likes:long,
    category_id:chararray, comments:long, channel_id:chararray);

data = FILTER records BY video_id != 'video_id';
grouped = GROUP data BY category_id;

top10_by_category = FOREACH grouped {
    ordered = ORDER data BY likes DESC;
    limited = LIMIT ordered 10;
    GENERATE group AS category_id,
        FLATTEN(limited.(video_id, title, channel_title, likes));
};
DUMP top10_by_category;
STORE top10_by_category
INTO '/user/DELL/youtube_project/output/pig/top10_liked_by_category'
USING PigStorage('\t');

```

Execution

```
pig -x mapreduce C:\FinalProject\PigAnalysis\scripts\top10LikedByCategories.pig
```

```

2026-09-10 17:02:12,098 [main] INFO org.apache.hadoop.yarn.client.RMPProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-10 17:02:12,107 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED.
Redirecting to job history server
2026-09-10 17:02:12,129 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Encountered Warning FIELD_DISCARDED_TYPE_CONVERSION_FAILED 3 time(s).
2026-09-10 17:02:12,129 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-10 17:02:12,143 [main] INFO org.apache.pig.Main - Pig script completed in 1 minute, 6 seconds and 383 milliseconds (66383 ms)

```

Output

Output fields: category_id, video_id, title, channel_title, likes

Local text output: C:\FinalProject\PigAnalysis\outputFiles\top10_liked_by_category.txt

hdfs dfs -cat /user/DELL/youtube_project/output/pig/top10_liked_by_category/part-r-00000

```

C:\Windows\System32>type C:\FinalProject\PigAnalysis\outputFiles\top10_liked_by_category.txt
category_id video_id title channel_title likes
1 G-7U-FDq1IA Gorillaz - PAC-MAN ft. Schoolboy Q (Episode Five) Gorillaz 777552
1 ZLKZKmdZEjM Gorillaz - D[-sol]- ft. Fatoumata Diawara (Episode Two) Gorillaz 631935
1 22N2EDRkPas For $1,000,000, Which House Would You Pick =fñö NYKentertain 597885
1 QTt7301PR5k Gorillaz - Momentary Bliss ft. slowthai & Slaves (Episode One) Gorillaz 576405
1 QB5tC0ie83Q =fæö=fu||nñÄ Parents Finally Get Alone TimefC³ Then Their Kid TRAPS Them in a City Scavenger Hunt =fyj|=fÄj FrameRush 562619
1 CJ68kQLS250 Gorillaz - The Pink Phantom ft. Elton John & 6LACK (Episode Seven) Gorillaz 551872
1 FJVV8o6vgso Gorillaz - The Lost Chord ft. Lee John (Episode Nine) Gorillaz 480430
1 PKXl0FW_ZCA Gorillaz - Aries ft. Peter Hook & Georgia (Episode Three) Gorillaz 478874
1 JRWdjIPGcmJ Fry Is Stuck in an Endless Loop. #futurama #funny Shiny Uploads 446968
1 lkvLY-iTb-g This is the only superpower that is actually achievable in real life.#film #movie Scene Analysis 425268
2 ziOUkBoTORE He Backed Into Him on PURPOSE! =fyj #dashcam #shorts Memento 313165
2 nkzqaVFjUeo This will never not be funny =fyë=fyë=fyë #comedy #funny #reaction Vibes On Vibes 311786
2 JnEf7N0fFKA Do not brake check semi trucks #trucking #truckdriver #pickuptruck Alan Trucker 289019
2 V-GUEFTqBHK Karen Turns Infront Of Biker Glint 197056
2 Fqmp9CXt3tk What year are you in? The evolution of fixing a flat tire! #car #Carlovers Car Lover's Corner 195346
2 uxyeAFHvQWg Biker Can't Handel The Heat =fyj 2WheelChaos 194325
2 6FKVVoPVH9U FERRARI STOPPED ME REBUILDING MY WRECKED 296 GTB Mat Armstrong 192918
2 J0qPPM30Vcw How to repair windshield rock chip #autoglass Ayfoos Its Otis 183738

```

```

C:\Windows\System32>hdfs dfs -cat /user/DELL/youtube_project/output/pig/top10_liked_by_category/part-r-00000
1 G-7U-FDq1IA Gorillaz - PAC-MAN ft. Schoolboy Q (Episode Five) Gorillaz 777552
1 ZLKZKmdZEjM Gorillaz - D[-sol]- ft. Fatoumata Diawara (Episode Two) Gorillaz 631935
1 22N2EDRkPas For $1,000,000, Which House Would You Pick =fñö NYKentertain 597885
1 QTt7301PR5k Gorillaz - Momentary Bliss ft. slowthai & Slaves (Episode One) Gorillaz 576405
1 QB5tC0ie83Q =fæö=fu||nñÄ Parents Finally Get Alone TimefC³ Then Their Kid TRAPS Them in a City Scavenger Hunt =fyj|=fÄj FrameRush 562619
1 CJ68kQLS250 Gorillaz - The Pink Phantom ft. Elton John & 6LACK (Episode Seven) Gorillaz 551872
1 FJVV8o6vgso Gorillaz - The Lost Chord ft. Lee John (Episode Nine) Gorillaz 480430
1 PKXl0FW_ZCA Gorillaz - Aries ft. Peter Hook & Georgia (Episode Three) Gorillaz 478874
1 JRWdjIPGcmJ Fry Is Stuck in an Endless Loop. #futurama #funny Shiny Uploads 446968
1 lkvLY-iTb-g This is the only superpower that is actually achievable in real life.#film #movie Scene Analysis 425268
2 ziOUkBoTORE He Backed Into Him on PURPOSE! =fyj #dashcam #shorts Memento 313165
2 nkzqaVFjUeo This will never not be funny =fyë=fyë=fyë #comedy #funny #reaction Vibes On Vibes 311786
2 JnEf7N0fFKA Do not brake check semi trucks #trucking #truckdriver #pickuptruck Alan Trucker 289019
2 V-GUEFTqBHK Karen Turns Infront Of Biker Glint 197056
2 Fqmp9CXt3tk What year are you in? The evolution of fixing a flat tire! #car #Carlovers Car Lover's Corner 195346
2 uxyeAFHvQWg Biker Can't Handel The Heat =fyj 2WheelChaos 194325
2 6FKVVoPVH9U FERRARI STOPPED ME REBUILDING MY WRECKED 296 GTB Mat Armstrong 192918
2 J0qPPM30Vcw How to repair windshield rock chip #autoglass Ayfoos Its Otis 183738
2 xz2_-7rH-jQ ICEtow Truck Takes The Shortcut!C³ Through The Grass =fyj|CvY #towtruck #repo #mercedes #shorts Miami Repo Show 169337
2 APTEBI6djzc Trolling a Road Rager =fñü HandsOnTheWheel 160853
2 J0qPPM30Vcw How to repair windshield rock chip #autoglass Ayfoos Its Otis 183738

```

4) Top 10 Viewed YouTube Videos

Returns the ten videos with the highest number of views in descending order.

Script

```

records = LOAD '/user/DELL/youtube_project/clean/US_Trending_Clean.tsv'
USING PigStorage('\t')
AS (video_id:chararray, trending_date:chararray, title:chararray,
    channel_title:chararray, views:long, likes:long,
    category_id:chararray, comments:long, channel_id:chararray);

data = FILTER records BY video_id != 'video_id';
selected = FOREACH data GENERATE video_id, title, channel_title, views;
sorted = ORDER selected BY views DESC;
top10 = LIMIT sorted 10;
DUMP top10;
STORE top10 INTO '/user/DELL/youtube_project/output/pig/top10_viewed'
USING PigStorage('\t');

```

Execution

```
pig -x mapreduce C:\FinalProject\PigAnalysis\scripts\top10Viewed.pig
```

```
2026-09-10 17:12:27,917 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-10 17:12:27,922 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-10 17:12:27,935 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Encountered Warning FIELD_DISCARDED_TYPE_CONVERSION_FAILED 1 time(s).
2026-09-10 17:12:27,942 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-10 17:12:27,947 [main] INFO org.apache.pig.Main - Pig script completed in 3 minutes, 29 seconds and 130 milliseconds (20913
```

Output

Output fields: video_id, title, channel_title, views

Local text output: C:\FinalProject\PigAnalysis\outputFiles\top10_viewed.txt

hdfs dfs -cat /user/DELL/youtube_project/output/pig/top10_viewed/part-r-00000

```
C:\Windows\System32>type C:\FinalProject\PigAnalysis\outputFiles\top10_viewed.txt
video_id title channel_title views
HyHNuVaZJ-k Gorillaz - Feel Good Inc. (Official Video) Gorillaz 1105355210
1V_xRb0x9aw Gorillaz - Clint Eastwood (Official Video) Gorillaz 1014925454
qdsTUfDTEHQ T.I. - No Mediocre ft. Iggy Azalea TIVEVO 420729382
etfIdtm-OC8 T.I. - About The Money ft. Young Thug TIVEVO 387242997
uAOR6ib95KQ Gorillaz - DARE (Official Video) Gorillaz 367587921
koVHN6e04Xg T.I. - Live Your Life ft. Rihanna [Official Video] T.I. 363500509
Caryj0dYFa0 Brawl Stars: No Time to Explain Brawl Stars 318547842
04mfKJWDSzI Gorillaz - On Melancholy Hill (Official Video) Gorillaz 300678427
nQJACVmanKY T.I. - Whatever You Like [Official Video] T.I. 274160249
0c0sLTMSl9c Brawl Stars Animation: Piper's Sugar & Spice! Brawl Stars 258757017
```

```
C:\Windows\System32>hdfs dfs -cat /user/DELL/youtube_project/output/pig/top10_viewed/part-r-00000
HyHNuVaZJ-k Gorillaz - Feel Good Inc. (Official Video) Gorillaz 1105355210
1V_xRb0x9aw Gorillaz - Clint Eastwood (Official Video) Gorillaz 1014925454
qdsTUfDTEHQ T.I. - No Mediocre ft. Iggy Azalea TIVEVO 420729382
etfIdtm-OC8 T.I. - About The Money ft. Young Thug TIVEVO 387242997
uAOR6ib95KQ Gorillaz - DARE (Official Video) Gorillaz 367587921
koVHN6e04Xg T.I. - Live Your Life ft. Rihanna [Official Video] T.I. 363500509
Caryj0dYFa0 Brawl Stars: No Time to Explain Brawl Stars 318547842
04mfKJWDSzI Gorillaz - On Melancholy Hill (Official Video) Gorillaz 300678427
nQJACVmanKY T.I. - Whatever You Like [Official Video] T.I. 274160249
0c0sLTMSl9c Brawl Stars Animation: Piper's Sugar & Spice! Brawl Stars 258757017
```

5) Top 10 Viewed Videos by Categories

Returns up to ten highest-viewed videos for each category. The data is grouped by category_id; records inside each category are sorted by views in descending order and limited to ten.

Script

```
records = LOAD '/user/DELL/youtube_project/clean/US_Trending_Clean.tsv'
USING PigStorage('\t')
AS (video_id:chararray, trending_date:chararray, title:chararray,
    channel_title:chararray, views:long, likes:long,
    category_id:chararray, comments:long, channel_id:chararray);

data = FILTER records BY video_id != 'video_id';
grouped = GROUP data BY category_id;

top10_by_category = FOREACH grouped {
    ordered = ORDER data BY views DESC;
    limited = LIMIT ordered 10;
    GENERATE group AS category_id,
        FLATTEN(limited.(video_id, title, channel_title, views));
};
DUMP top10_by_category;
STORE top10_by_category
INTO '/user/DELL/youtube_project/output/pig/top10_viewed_by_category'
USING PigStorage('\t');
```

Execution

```
pig -x mapreduce C:\FinalProject\PigAnalysis\scripts\top10ViewedByCategories.pig
```

```
2026-09-10 17:15:31,080 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-10 17:15:31,084 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicati
onStatus=SUCCEEDED. Redirecting to job history server
2026-09-10 17:15:31,102 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-10 17:15:31,106 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicati
onStatus=SUCCEEDED. Redirecting to job history server
2026-09-10 17:15:31,128 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-10 17:15:31,134 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicati
onStatus=SUCCEEDED. Redirecting to job history server
2026-09-10 17:15:31,153 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Encountered Wa
rning FIELD_DISCARDED_TYPE_CONVERSION_FAILED 3 time(s).
2026-09-10 17:15:31,153 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-10 17:15:31,170 [main] INFO org.apache.pig.Main - Pig script completed in 51 seconds and 443 milliseconds (51443 ms)
```

Output

Output fields: category_id, video_id, title, channel_title, views

```
hdfs dfs -cat /user/DELL/youtube_project/output/pig/top10_viewed_by_category/part-r-00000
```

Local text output: C:\FinalProject\PigAnalysis\outputFiles\top10_viewed_by_category.txt

```
C:\Windows\System32>hdfs dfs -cat /user/DELL/youtube_project/output/pig/top10_viewed_by_category/part-r-00000
1 UMMzKXsY9A4 Final Destination Bloodlines | Official Trailer Warner Bros. 41958999
1 IhgcUAR03Uo THE BRIDE! | Official Trailer Warner Bros. 40586575
1 G-7U-FDq1IA Gorillaz - PAC-MAN ft. Schoolboy Q (Episode Five) Gorillaz 38314234
1 ZLKZKmdZEjM Gorillaz - D|-sol|- ft. Fatoumata Diawara (Episode Two) Gorillaz 36879088
1 OpThntO9ixc Weapons | Official Trailer Warner Bros. 33007071
1 FSAz556s0FM The Conjuring: Last Rites | Official Trailer Warner Bros. 27939160
1 3FLCdIYShEQ 'Muthering Heights' | Official Trailer Warner Bros. 27693998
1 bMgfsdYoEEo The Conjuring: Last Rites | Official Trailer Warner Bros. 26144243
1 QKHYSFXqN0I Weapons | Official Trailer 2 Warner Bros. 25060112
1 QTT7301PR5k Gorillaz - Momentary Bliss ft. slowthai & Slaves (Episode One) Gorillaz 24111947
2 Fqmp9CXT3tk What year are you in? The evolution of fixing a flat tire! #car #Carlovers Car Lover's Corner 10532298
2 xz2_-7rH-jQ fCEIow Truck Takes The ShortcutfC9 Through The Grass #fy|fCY #towtruck #repo #mercedes #shorts Miami Repo Show 9815502
2 JnEf7N0fFKA Do not brake check semi trucks #trucking #truckdriver #pickuptruck Alan Trucker 7696009
2 ziOUkBoTORE He Backed Into Him on PURPOSE! #fy #dashcam #shorts Momento 7057150
2 Gixq8b2H5FM Comprehensive: 'Mercedes W210 E-Class Android Head Unit Upgrade & Full Installation' Best Mechanics Around the World 7044856
2 nkzqaVFJueo This will never not be funny #fy#fy#fy#fy #comedy #funny #reaction Vibes On Vibes 6854878
2 Oh7K9CjX19E #bike #street #highway #car #driver #power #autobahn #supercars #speed #drifting #4x4 #4wd #offroad Autobahn Performance 6347894
2 zjNlsoy6lrY S1000RR Aplastada y persecuci|n #automobile #karma #moto #s1000rr #humor Camillo Road City 5908639
2 6fKVoPvH9U FERRARI STOPPED ME REBUILDING MY WRECKED 296 GTB Mat Armstrong 5411080
2 J0qPPM30Vcw How to repair windshield rock chip #autoglass Ayfoo Its_Otis 5122817
10 HyHNuVaZJ-k Gorillaz - Feel Good Inc. (Official Video) Gorillaz 1105355210
```

```
C:\Windows\System32>type C:\FinalProject\PigAnalysis\outputFiles\top10_viewed_by_category.txt
category_id video_id title channel_title views
1 UMMzKXsY9A4 Final Destination Bloodlines | Official Trailer Warner Bros. 41958999
1 IhgcUAR03Uo THE BRIDE! | Official Trailer Warner Bros. 40586575
1 G-7U-FDq1IA Gorillaz - PAC-MAN ft. Schoolboy Q (Episode Five) Gorillaz 38314234
1 ZLKZKmdZEjM Gorillaz - D|-sol|- ft. Fatoumata Diawara (Episode Two) Gorillaz 36879088
1 OpThntO9ixc Weapons | Official Trailer Warner Bros. 33007071
1 FSAz556s0FM The Conjuring: Last Rites | Official Trailer Warner Bros. 27939160
1 3FLCdIYShEQ 'Muthering Heights' | Official Trailer Warner Bros. 27693998
1 bMgfsdYoEEo The Conjuring: Last Rites | Official Trailer Warner Bros. 26144243
1 QKHYSFXqN0I Weapons | Official Trailer 2 Warner Bros. 25060112
1 QTT7301PR5k Gorillaz - Momentary Bliss ft. slowthai & Slaves (Episode One) Gorillaz 24111947
2 Fqmp9CXT3tk What year are you in? The evolution of fixing a flat tire! #car #Carlovers Car Lover's Corner 10532298
2 xz2_-7rH-jQ fCEIow Truck Takes The ShortcutfC9 Through The Grass #fy|fCY #towtruck #repo #mercedes #shorts Miami Repo Show 9815502
2 JnEf7N0fFKA Do not brake check semi trucks #trucking #truckdriver #pickuptruck Alan Trucker 7696009
2 ziOUkBoTORE He Backed Into Him on PURPOSE! #fy #dashcam #shorts Momento 7057150
2 Gixq8b2H5FM Comprehensive: 'Mercedes W210 E-Class Android Head Unit Upgrade & Full Installation' Best Mechanics Around the World 7044856
2 nkzqaVFJueo This will never not be funny #fy#fy#fy#fy #comedy #funny #reaction Vibes On Vibes 6854878
2 Oh7K9CjX19E #bike #street #highway #car #driver #power #autobahn #supercars #speed #drifting #4x4 #4wd #offroad Autobahn Performance 6347894
2 zjNlsoy6lrY S1000RR Aplastada y persecuci|n #automobile #karma #moto #s1000rr #humor Camillo Road City 5908639
2 6fKVoPvH9U FERRARI STOPPED ME REBUILDING MY WRECKED 296 GTB Mat Armstrong 5411080
2 J0qPPM30Vcw How to repair windshield rock chip #autoglass Ayfoo Its_Otis 5122817
10 HyHNuVaZJ-k Gorillaz - Feel Good Inc. (Official Video) Gorillaz 1105355210
10 1V_xR0x09aw Gorillaz - Clint Eastwood (Official Video) Gorillaz 1014925454
10 qdsTUDfTEhQ T.I. - No Mediocre ft. Iggy Azalea TIVEVO 42729382
10 etfIdtm-OC8 T.I. - About The Money ft. Young Thug TIVEVO 387242997
10 uAOR6ib95kQ Gorillaz - DARE (Official Video) Gorillaz 367587921
10 koVlNge04Xg T.I. - Live Your Life ft. Rihanna [Official Video] T.I. 363500509
10 04mFKWDSZT Gorillaz - On Melancholy Hill (Official Video) Gorillaz 308678427
```

GitHub Repository

The completed project files were organized and pushed to a public GitHub repository. The repository follows a structure similar to the reference project while keeping the current project's Java preprocessing, three MapReduce analyses, five Pig analyses, datasets, and output files clearly separated.

Repository URL: <https://github.com/25tarek/YouTube-Data-Analysis-using-Hadoop>

Repository Branch: main

Visibility: Public

Repository Item	Purpose
Preprocessing/	Java CSV preprocessing project and Maven configuration.
Top10Categories/	MapReduce analysis for category-level video counts and Top 10 ranking.
Top10ViewedVideos/	MapReduce analysis for the Top 10 videos by views.
Top10Channels/	MapReduce analysis for the Top 10 channels by number of videos.
PigAnalysis/	Five Pig scripts and their local text output files.
MR Results/outputFiles/	Centralized text outputs produced by the MapReduce analyses.
mergedataset/	Dataset source information, raw US CSV, processed TSV, and data dictionary.
screenshots/	Folder reserved for report and execution evidence screenshots.

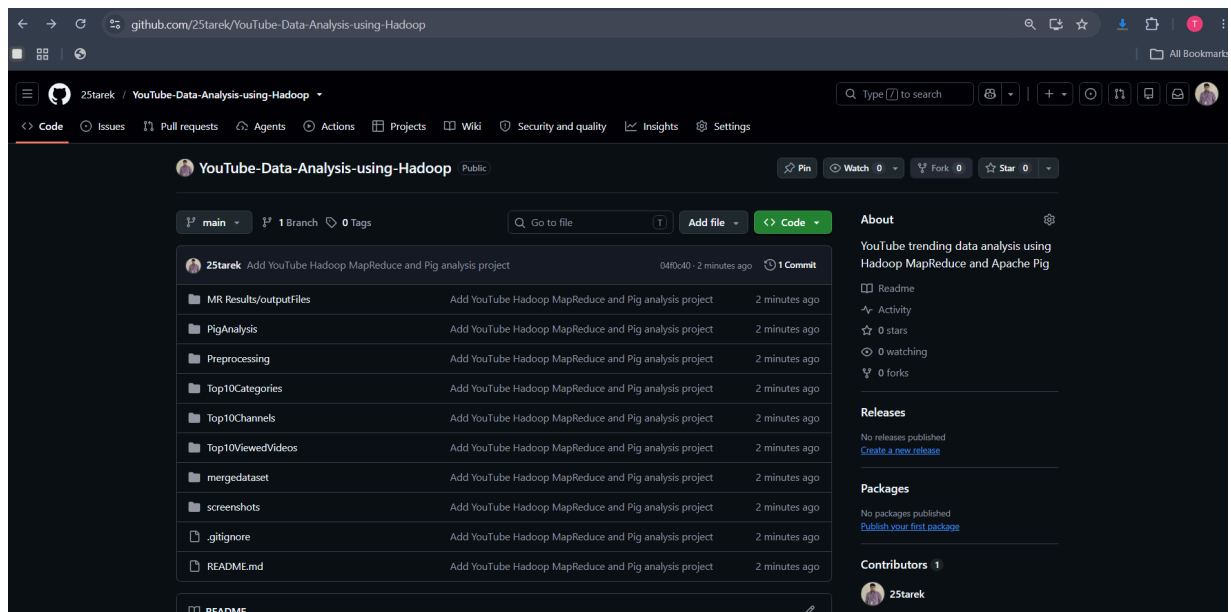


Figure: GitHub repository after the project source code and results were pushed.

Conclusion

This project demonstrates an end-to-end YouTube trending video data analysis workflow using Java, Hadoop HDFS, YARN MapReduce, and Apache Pig. The raw US trending dataset was preprocessed into a tab-separated file so that the same cleaned input could be used consistently by both MapReduce and Pig.

Three MapReduce analyses were implemented to identify the Top 10 categories by video count, the Top 10 most viewed videos, and the Top 10 channels by number of videos. Five Pig analyses were implemented for category counts, liked-video rankings, category-wise liked-video rankings, viewed-video rankings, and category-wise viewed-video rankings. The overlapping category and view-based analyses provide a useful cross-check between the Java MapReduce and Pig workflows.

The project source code, scripts, processed data, and result files were organized in a public GitHub repository to make the work easy to review and reproduce.

References

1. Kaggle dataset - YouTube Trending Videos Stats 2026:
<https://www.kaggle.com/datasets/bsthere/youtube-trending-videos-stats-2026>
2. Project GitHub repository: <https://github.com/25tarek/YouTube-Data-Analysis-using-Hadoop>
3. Reference project used for report structure:
<https://github.com/SiddhiPrabhu1995/YouTube-Data-Analysis-using-Hadoop>
4. Apache Hadoop documentation: <https://hadoop.apache.org/docs/r3.2.4/>
5. Apache Pig documentation: <https://pig.apache.org/docs/r0.18.0/>